

CC-OCR: A Comprehensive and Challenging OCR Benchmark for Evaluating Large Multimodal Models in Literacy

Zhibo Yang^{1,2*} Jun Tang^{2*} Zhaohai Li^{2*} Pengfei Wang^{2*} Jianqiang Wan^{2*} Humen Zhong^{2*}
Xuejing Liu² Mingkun Yang² Peng Wang² Shuai Bai² Lianwen Jin³ Junyang Lin^{2†}

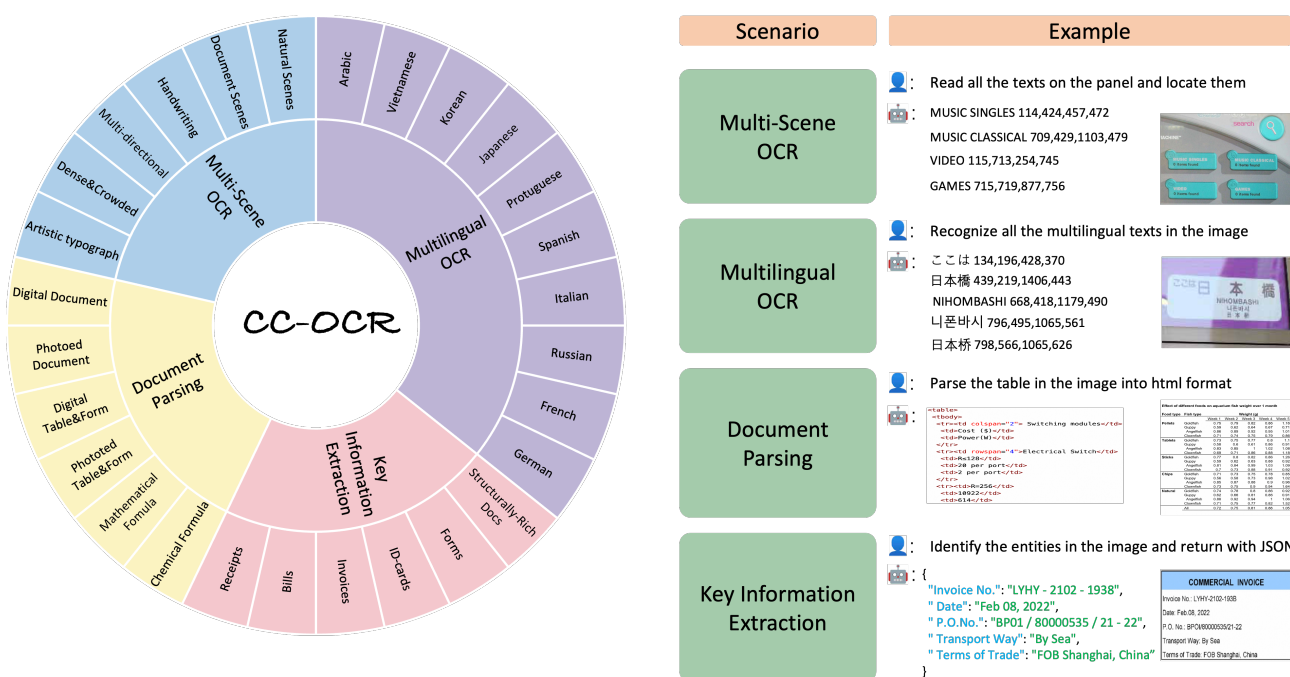
¹Huazhong University of Science and Technology ²Alibaba Group³ South China University of Technology

Figure 1. Task Taxonomy of CC-OCR.

Abstract

*Large Multimodal Models (LMMs) have demonstrated impressive performance in recognizing document images with natural language instructions. However, it remains unclear to what extent capabilities in literacy with rich structure and fine-grained visual challenges. The current landscape lacks a comprehensive benchmark to effectively measure the literate capabilities of LMMs. Existing benchmarks are often limited by narrow scenarios and specified tasks. To this end, we introduce **CC-OCR**, a comprehensive benchmark that possesses a diverse range of scenarios, tasks, and challenges.*

CC-OCR comprises four OCR-centric tracks: multi-scene text reading, multilingual text reading, document parsing, and key information extraction. It includes 39 subsets with 7,058 full annotated images, of which 41% are sourced from real applications, and released for the first time. We evaluate fourteen prominent LMMs and two well-known specialists to reveal both the strengths and weaknesses of these models, particularly in text grounding, multi-orientation, and hallucination of repetition. **CC-OCR** aims to comprehensively evaluate the capabilities of LMMs on OCR-centered tasks, facilitating continued progress in this crucial area. The benchmark can be accessed at the following link: [GitHub](#).

*Equal Contribution.

†Corresponding Author.

1. Introduction

Literate capability is crucial in today’s LMMs [3, 4, 8, 39], as it empowers models to engage with texts in images and multimodal dialogues critically. The ability to literate not only includes reading plain texts but also involves position grounding, structured layouts, and handling formatted input and structured output. LMMs hold great potential to unify these literate capabilities using natural language instructions and images as prompts, which meanwhile leads to heightened demand for innovative and comprehensive evaluations. In the OCR domain, specialized benchmarks exist to evaluate various downstream tasks such as text line recognition [15, 30], text detection [9, 10], formula recognition [27, 48], multilingual recognition [31], and table structure recognition [19, 23]. However, these benchmarks are not applicable for evaluating LMMs due to they are fragmented, purpose-diverse, and inconsistently annotated. Recent benchmarks like TextVQA [35] and DocVQA [29] aim to evaluate textual comprehension capabilities. Nevertheless, these benchmarks fall short of thoroughly and quantitatively evaluating their perceptual and interpretative literacy. OCR-Bench [21] is a benchmark for evaluating the line-level text recognition capabilities of LMMs, but it does not capture the broader full-image challenges.

To bridge these critical gaps, we introduce CC-OCR, a comprehensive and challenging benchmark designed specifically to evaluate LMMs across different dimensions of literacy capabilities, including multilingual text reading, text spotting, layout-aware document parsing, and key information extraction. These capabilities can pave the way for LMMs with a pathway for real-world applications, including document digitization, office automation, and urban monitoring.

CC-OCR is distinguished by two key features: the comprehensiveness of tasks and the challenges across various scenarios. On the one hand, from an application perspective, as presented in Fig. 1, we focus on four OCR-centric tasks. On the other hand, to highlight challenges, as shown in Fig. 2, CC-OCR covers fine-grained visual challenges (i.e., orientation-sensitivity, grounding, natural noise, and artistic text), decoding of various expressions, and structured inputs and outputs.

For thorough evaluation, we selected ten generalist models, including Cambrian-34B [40], Phi-4-Multimodal [2], Llama-4-maverick, GPT-4o [3], Gemini-1.5 [39], Claude-3.5-Sonnet [1], Qwen2-VL-72B [42], Qwen2.5-VL-72B [5], InternVL2 [8] and InternVL3 [53]. We also test four specialist models including KOSMOS2.5 [26], TextMonkey [22], Florence [44], GOT [43], and two OCR APIs including PadleOCR, GoogleOCR.

Our comprehensive evaluation reveals that although LMMs achieve remarkable performance on high-level scene understanding tasks like full-text captioning, current LMMs

decrease to deliver satisfactory performance in challenging text-rich images. Qwen2.5-VL-72B and Gemini-1.5-Pro are the two top-performing models. Qwen2.5-VL-72B achieves first in the multi-scene OCR, multilingual OCR, parsing, and KIE tracks, while Gemini-1.5-Pro takes second place in multi-scene OCR, multilingual OCR, and parsing. Our findings also reveal several interesting insights: (1) Text in natural scenes remains underexplored, with performance being over 15% lower than those for documents. (2) Structured formats are challenging to recognize, as evidenced by the poorer performance in parsing and KIE tasks compared to text-only recognition. (3) Most models exhibit weaker multilingual capabilities compared to their proficiency in Chinese and English. (4) Fine-grained text grounding is a common limitation across all models.

To summarize, our major contributions are as follows:

- We introduce CC-OCR, a comprehensive and challenging benchmark tailored to OCR, encompassing four primary tasks and 39 sub-tasks, featuring multi-scene, multilingual, and multi-format output.
- We conduct a thorough evaluation of ten different LMMs, including five generalist LMMs and four specialist LMMs designed for document, providing a quantitative analysis of their strengths and weaknesses.
- Our evaluation reveals valuable insights guidance for enhancing LMMs to achieve advanced levels of literate intelligence.

2. Related Work

Multi-Scene Text Reading. There are a variety of specialized benchmarks in the early era of deep learning, such as natural-scene [15, 34], web-scene [10], multi-directional and curved text [49]. Current OCRBench [21] for evaluating LMMs mainly focuses on line-granularity recognition. DocLocal4K [11] and FOX [20] mainly curate data from document images. KOSMOS-2.5 [26] proposes two benchmarks: OCREval and MarkdownEval, but both lack data from natural scenes. These benchmarks focus on a specific challenge and therefore fail to serve as comprehensive benchmarks for evaluating general-purpose LMMs.

Multilingual Text Reading emphasizes the capability of text recognition across various languages. Related multilingual evaluation datasets are scarce, especially those for whole images. MLT 2019 [31], widely used for multilingual text reading, provides a diverse set of images with text in multiple languages. MTVQA [37] integrates visual and textual reasoning through question-answering (QA) pairs, adding complexity to the evaluation. However, it lacks comprehensive OCR annotations, focusing mainly on QA pairs, which limits its utility for OCR tasks.

Document Parsing encompasses a diverse range of scenarios and tasks, including formula recognition [6], table [50] and chart analysis [7], element detection [51], and layout

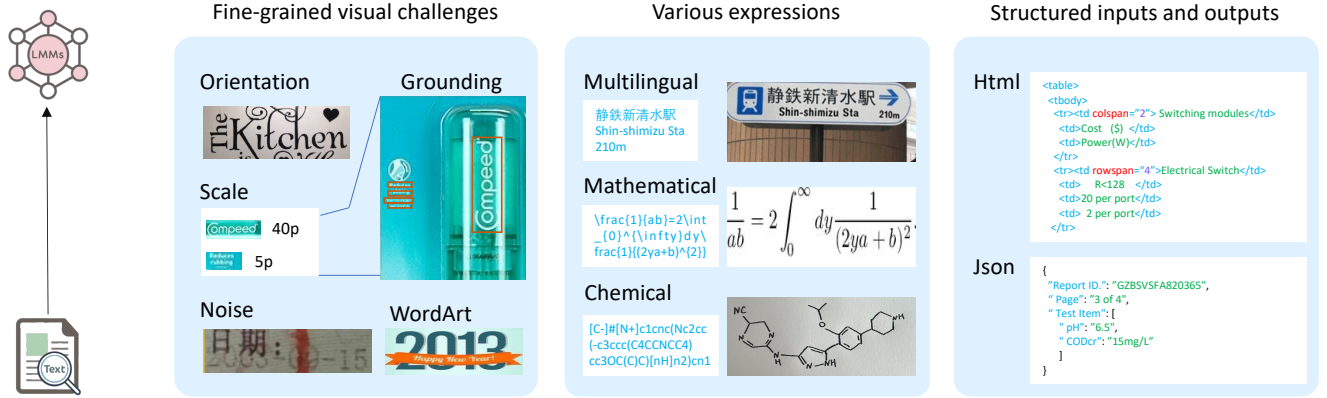


Figure 2. Challenge Taxonomy of CC-OCR.

analysis [25, 38]. Most LMMs evaluate their recognition capabilities exclusively on scanned documents, overlooking real-world documents. The factors in real-shot documents pose significant challenges for text content and structural recognition and limit the robustness of current models.

Key Information Extraction. Currently, the KIE data can be divided into two categories, constrained-category and open-category datasets, depending on whether the key to be extracted is predefined. For the constrained KIE, the target categories to be extracted are given. There are public datasets available, such as SROIE [12], CORD, EPHOIE [41], and POIE [17]. These benchmarks also provide end-to-end representation (i.e. dictionary form) which can be adopted to evaluate the large models. SIBR [45] and HUST-CELL [47] are typical open-category benchmarks, the structures of which are represented by entities and various types of entity links. Therefore, this annotation format cannot be directly used to evaluate large models.

3. Dataset

3.1. Dataset Collection

The CC-OCR dataset is constructed following three criteria: diversity, practicality, and challenge. In alignment with these criteria, we prioritize using natural scene images, real-world captured documents, and handwritten images. The data sources consist of three parts: existing benchmarks with qualified annotations, re-annotated datasets meeting LMM evaluation standards, and newly-collected data. The detailed composition of CC-OCR is presented in Appendix.

Multi-Scene OCR benchmark primarily comprises data in both English and Chinese. The English segment leverages established academic benchmark datasets, which encompass natural scene (TotalText [9], IC15 [15], HierAgent), documents (FUNSD [14], CORD [32], IAM [28]), multi-orientation (InverseText [46]), and artistic text (LAION-OCR [33]). In contrast, the Chinese component consists entirely of newly collected datasets, encompassing street

scene (Scene-zh), web scene (UGC-ver-zh, DAST-zh [36]), and document (Doc-zh, Doc-hw-zh). HierAgent is a subset of Hiertext [24], consisting of data related to mobile devices, computer screen-shots, and machine control interfaces. The LAION-OCR includes 400 randomly selected and re-annotated images from LAION-5B [33]. In total, this sub-benchmark contains 1850 natural scene images, 350 document images, and 550 web images.

Multilingual OCR. In this sub-benchmark, each language consists of 10 real-shot document images and 140 natural scene images. This ratio is strategically designed to achieve a balance in the number of words. The document images are entirely newly-collected, featuring multi-orientation and real-shot. The natural scene images are partially derived from the re-annotation of the MTVQA [37] and partially sourced from newly-collected data, particularly for Russian, Spanish, and Portuguese.

Document Parsing benchmark includes three types of data: 300 document images, 300 table images, and 200 formula images. All data is sourced from in-house collections. Table and formula datasets refer to those documents where the main content is composed of tables or formulas. In contrast, the dataset of *documents* is primarily made up of plain texts, with only a small portion of the areas containing tables and formulas.

Key Information Extraction benchmark can be divided into two types: constrained-category with 1008 images and open-category with 1000 images. For the constrained-category, data are derived from SROIE, CORD, EPHOIE, and POIE. For balance, we sample the 250 samples from POIE. Open-category images are derived from SIBR and HUST-CELL. However, the original annotation of these two datasets designed to evaluate entities and entity relationships is no longer applicable. We re-annotated these two datasets using a standardized JSON format to maintain consistency with the previously constrained-category datasets. The newly annotated open-category dataset is renamed as **SOLD**, short

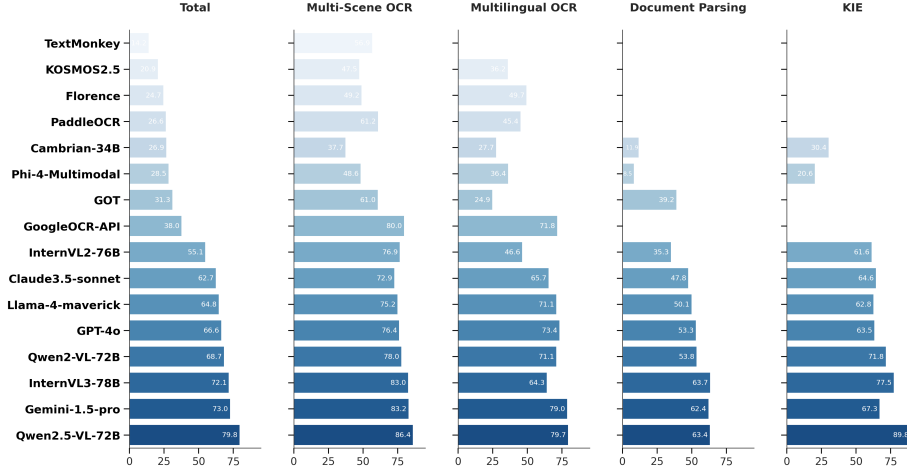


Figure 3. Comparative Analysis of Large Models on Four Tracks and an Overall Performance.

for Structurally-rich Open Layout Dataset.

3.2. Dataset Comparison

The evaluation of Large Multimodal Models (LMMs) for OCR tasks requires comprehensive benchmarks that can assess various capabilities. While numerous OCR datasets emerged during the deep learning era, they were typically designed for specific tasks rather than holistic evaluation. To position our work, we compare CC-OCR with four recent contemporary benchmarks (OCRBench, Fox, DocLocal4K, and KOSMOS2.5-Eval), as detailed in Tab. 1.

One difference among these benchmarks lies in their text processing granularity. Most existing benchmarks focus on full-image analysis, with OCRBench being the exception in its emphasis on line-level recognition and single-question VQA tasks. This granular approach, while valuable for specific applications, limits the comprehensive evaluation of LMM capabilities. Language coverage represents another significant distinguishing factor. DocLocal4K and KOSMOS2.5-Eval are limited to English-only evaluation, while OCRBench and Fox extend their coverage to include both English and Chinese. CC-OCR substantially broadens this linguistic scope by supporting ten major languages spanning multiple language families. These include Indo-European languages (English, German, French, Spanish, Portuguese, Russian), East Asian languages (Chinese, Japanese, Korean), Arabic from the Afroasiatic family, and Vietnamese from the Austroasiatic family. The benchmarks also differ markedly in their coverage of real-world scenarios. While most existing benchmarks lack support for one or more key scenarios, OCRBench and CC-OCR encompass all four essential scenarios. In terms of evaluation capabilities, most benchmarks focus solely on OCR performance. KOSMOS2.5-Eval extends this by including both OCR and parsing tracks. Our CC-OCR further expands the evaluation scope by incorporating three distinct tracks: OCR performance,

document parsing, and key information extraction. In general, the proposed CC-OCR benchmark distinguishes itself by offering the most comprehensive evaluation framework among existing benchmarks. Its multi-dimensional coverage across languages, scenarios, and evaluation tracks enables a thorough assessment of LMM capabilities in diverse real-world applications.

3.3. Dataset Annotation

Based on the categorization of elements and structures present in the images, alongside the intended output objectives, we classify the annotations into four distinct types: texts, structured documents, tables, and information extraction. An iterative annotation refinement process is employed for all new annotations. The process begins with model-based pre-annotation, proceeds with cross-validation by different models, and culminates in manual correction to ensure annotation accuracy.

Annotation of Texts. The text annotations cover multi-scene and multilingual OCR, for which we provide text transcriptions and word-level grounding (line-level bounding boxes for languages such as Chinese and Japanese).

Annotation of Documents. For documents, due to their diverse elements, we use LaTeX format for representation. Firstly, we utilize the Mathpix-Markdown-IT tool to convert image content into the LaTeX format. We retain the plain text, formulas, and tables, while style-related symbols are removed. For handwritten formulas and chemical molecular formulas, we first generate pseudo-labels using a recognition model and then manually correct any recognition errors. The annotations for these data types are formatted in LaTeX code and SMILES format, respectively. During evaluation, the accuracy of tables and formulas within the document sub-set is also assessed using edit distance.

Annotation of Structured Tables. For the evaluation of

Benchmark	Granularity	Language	Scene				Track			Num
			Doc	Natural Scene	Web	Handwritten	OCR	Parsing	KIE	
OCRBench [21]	Line Image	2	✓	✓	✓	✓	✓	✗	✓	1,000
FOX [20]	Full Image	2	✓	✓	✗	✗	✓	✗	✗	612
DocLocal4K [11]	Full Image	1	✓	✓	✓	✗	✓	✗	✗	4,250
KOSMOS2.5-Eval [26]	Full Image	1	✓	✗	✓	✗	✓	✓	✗	7,990
CC-OCR	Full Image	10	✓	✓	✓	✓	✓	✓	✓	7,058

Table 1. Comparisons of different benchmarks for evaluating LMMs.

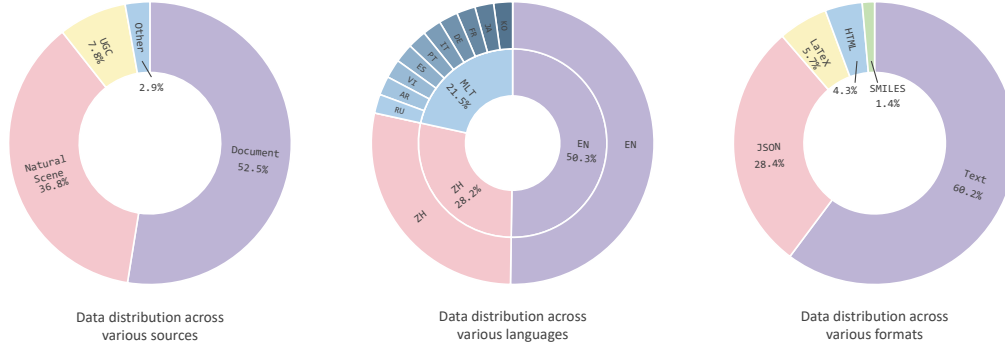


Figure 4. Statistics of Data Distribution.

table structures, we use the HTML format. This is because HTML can structurally represent and evaluate various complex structures, such as row and column spanning. HTML evaluation is applied to all table subsets of document parsing.

Annotation of Information Extraction. As mentioned in the data collection section, all extraction tasks are uniformly represented in JSON format. For the open-vocabulary and open-structured datasets, SOLD-SIBR and SOLD-CELL, we conduct a new round of annotation. We add annotations for hierarchical structures, table-format key-value (KV) groups, and items without keys. Finally, the annotation results are reviewed and verified using both OCR experts and large language models. For a detailed process, please refer to Appendix.

3.4. Dataset Analysis

The benchmark consists of four main tracks: Conventional OCR, Multilingual OCR, Document Parsing, and Key Information Extraction. All Chinese data in Multi-Scene OCR are newly collected. The multilingual data is composed equally of re-annotated open-source data and newly collected data. All document parsing data is newly collected. For KIE, the data of the constrained-category has been re-annotated. Detailed composition of the proposed CC-OCR benchmark is presented in Appendix.

To better demonstrate the balance of the dataset, we conduct a three-dimensional analysis of its distribution, with the results presented in Fig. 4. In terms of the source scenarios, 52.5% of the data are derived from documents, evenly divided between digital and real-shot documents. Natural

scene data accounts for 36.8%, while web images make up 7.8%. From a linguistic perspective, English accounts for the largest proportion at 50.3%, followed by Chinese at 28.2%. The remaining 10 languages are evenly distributed, collectively comprising the remaining 21.5%. Regarding data format, plain text constitutes 60.2% of the dataset, JSON format accounts for 28.4%, LaTeX for 5.7%, HTML for 4.3%, and 1.4% is the SMILES format for molecular formula.

4. Evaluation Method

For evaluating OCR, both text sequences and positional sequences, we employ evaluation metrics similar to those used in TextMonkey [22], namely Eval-Trans and Eval-Pos. Given that the majority of large models exhibit limited grounding capabilities, we primarily use Eval-Trans in our main experiments. For document parsing and formula recognition, we employ the Normalized Edit Distance [18] (NED) for evaluation. For table parsing, we use Tree Edit Distance-based Similarity (TEDS) [52]. For Key information extraction, we follow previous work [13, 16] and use field-level F1 score for evaluation. Detailed evaluation methods can be found in Appendix.

5. Experiments

In this section, we evaluate ten representative LMMs either with open-source models or commercial APIs. The commercial APIs with specific versions are GPT-4o-2024-08-06, Gemini-1.5-Pro-002, Claude-3.5-Sonnet-20241022. The open-source LMMs include KOSMOS2.5, TextMonkey,

Method	Natural Scene					Document					Web UGC			Total
	TotalText	IC15	InverseText	HierAgent	Scene _{zh}	FUNSD	CORD	IAM	Doc _{zh}	Doc-hw _{zh}	LAION _{OCR}	UGC-ver _{zh}	DAST _{zh}	
PaddleOCR	37.03	41.53	22.68	58.6	72	66.84	81.1	27.68	88.33	67.37	59.56	87.6	84.96	61.18
GoogleOCR-API	66.29	56.28	60.24	80.43	66.28	89.21	90.6	87.47	92.37	85.49	82.87	90.84	92.02	80.03
KOSMOS2.5 [26]	37.51	36.46	16.18	49.93	11.94	90.34	90.56	90.51	59.09	13.69	66.67	13.14	42.17	47.55
Florence [44]	72.76	61.89	53.23	63.94	22.28	73.46	79.35	79.9	15.7	9.56	85.32	3.41	19.29	49.24
TextMonkey [22]	69.84	46.11	45.18	51.19	55.17	76.39	79.76	69.96	54.17	26.67	83.16	33.99	47.84	56.88
GOT [43]	36.2	46.1	20.94	47.07	65.23	83.87	66.98	65.33	75.56	63.17	62.7	82.34	77.57	61.00
Cambrian-34B [40]	69.52	52.31	50.25	13.51	13.67	28.04	68.03	85.78	6.17	7.00	78.92	10.43	76.8	37.66
Phi-4-Multimodal [2]	58.75	52.93	40.21	33.93	34.3	67.55	77.65	90.63	23.00	12.78	77.47	22.05	88.5	48.64
Claude-3.5-Sonnet [1]	76.09	48.1	39.67	66.64	53.62	85.73	92.32	96.02	82.33	66.09	85.51	80.04	75.14	72.87
Llama-4-maverick	79.57	64.59	72.63	62.23	52.75	86.22	92.19	93.87	84.65	53.96	85.46	75.47	89.1	75.23
GPT-4o[3]	79.82	61.52	77.03	61.85	58.3	86.08	86.46	95.23	77.96	61.72	86.45	87.64	73.12	76.40
InternVL2-76B [8]	80.15	66.39	72.04	62.18	80.01	78.69	82.14	91.31	79.91	70.65	85.62	86.74	64.18	76.92
Qwen2-VL-72B [42]	79.21	65.49	69.77	55.87	60.33	84.39	88.91	95.8	81.59	75.08	85.37	88.61	82.9	77.95
InternVL3-78B [53]	78.85	66.95	72.39	70.46	85.41	86.3	89.43	91.67	87.78	83.32	86.22	94.49	92.3	82.95
Gemini-1.5-Pro [39]	79.76	62.97	78.83	74.83	64.05	85.57	94.77	95.52	90.51	84.07	86.44	92.58	92.31	83.25
Qwen2.5-VL-72B [5]	79.85	66.86	75.5	82.16	82.66	87.6	93.53	96.19	93.34	90.39	87.32	95.23	92.6	86.40

Table 2. Results of LMMs on subsets of Multi-Scene OCR track. The upper part of the table refers to the specialist models, while the lower part refers to the generalist models.

Florence, GOT, Cambrian-34B, Phi-4-Multimodal, Llama-4-maverick, InternVL2-76B, InternVL3-78B, Qwen2-VL-72B and Qwen2.5-VL-72B.

The overall evaluation results are shown in Fig. 3. It can be observed that Qwen2.5-VL-72B achieves the highest ranking with an average score of 79.8, securing first place in all four tracks. This demonstrates its outstanding and robust OCR capabilities. Gemini-1.5-Pro ranks second overall, with an average score of 73.0. The results indicate that generalist models tend to perform better than specialized ones.

5.1. Multi-Scene OCR

We evaluate the generic LMMs and OCR-specialized LMMs on the track of multi-scene OCR. As presented in Tab. 2, generic LMMs models significantly outperform specialized models, primarily due to their huge parameter and training data. Among all the models, Qwen2.5-VL-72B performs the best with a score of 86.40, followed by Gemini-1.5-Pro with a score of 83.25. In the category of specialized models, GOT is the top performer with a score of 61.00. And for OCR APIs, GoogleOCR achieve impressive score of 80.03. Across different scenes, it is evident that the performance exhibits considerable variation. For most models, the average performance in natural scenes is approximately 15% lower compared to their results in document scenes.

5.2. Multilingual OCR

As shown in Tab. 3, the performance gap among models in the multilingual track is more notable. Among the generalist models, Qwen2.5-VL-72B ranks first with a score of 79.68. In the specialized group, Florence is the best and achieves a score of 49.70. And for OCR APIs, GoogleOCR achieve 71.78 on average. Additionally, the table illustrates a significant variation across different languages for every model. The accuracy of languages in Asia, such as Japanese, Korean,

Vietnamese, and Arabic, is generally lower than languages in the Latin family. Among the Asian languages, Japanese exhibits the lowest performance for most generalist models, which we attribute to the higher prevalence of vertical text in this typical language. In the Latin family, German and Italian often exhibit lower performance, possibly due to the presence of more special characters. Overall, the multilingual capabilities of most models are inferior to their performance in Chinese and English with the results presented in Tab. 2. Therefore, improving multilingual capabilities is a future direction for most models.

5.3. Document Parsing

In the document parsing track, most specialized models, except GOT do not support LaTeX or HTML formats. Therefore, we primarily evaluate the generalist models, and the results are presented in Tab. 4. InternVL3-78B, Qwen2.5-VL-72B demonstrate superior performance across various document parsing subsets, with InternVL3-78B achieving the best performance, scoring 63.66.

However, a detailed analysis of individual sub-sets reveals that the top-performing large models still attain average scores below 70. For instance, Qwen2.5-VL-72B achieves an average score of 60.49 in document content structuring and 76.49 in table recognition, indicating considerable room for improvement. This also demonstrates that our document parsing benchmark is comprehensive and challenging. More challenging document image examples can be found in Appendix.

Furthermore, we observe that most models perform better on English than Chinese documents. For example, Gemini-1.5-Pro scores 5% higher on English tables than on Chinese tables, and Qwen2.5-VL-72B performs 8% higher on English document than on Chinese document. The gap indicates that there is an imbalance in the preparation of training

Method	Korean	Japanese	Vietnamese	French	German	Italian	Spanish	Portuguese	Russian	Arabic	Total
PaddleOCR	62.41	60.14	15.36	44.29	45.95	48.91	42.68	43.59	10.03	80.4	45.38
GoogleOCR-API	85.32	77.46	63.15	73.4	64.8	67.67	67.93	69.72	57.69	90.62	71.78
GOT	27.53	43.66	7.92	33.74	28.73	28.53	31.63	25.20	3.85	18.72	24.95
KOSMOS2.5	26.95	26.94	27.24	48.93	47.04	44.59	52.92	54.56	11.13	22.04	36.23
Florence	27.93	34.76	21.96	70.21	61.30	61.62	70.44	67.78	26.74	54.24	49.70
Cambrian-34B	11.13	15.52	16.36	31.56	27.69	34.74	60.83	59.16	14.5	5.67	27.71
Phi-4-Multimodal	22.32	28.59	11.95	57.23	42.55	34	53.73	61.39	21.21	30.97	36.4
InternVL2-76B	31.58	44.52	20.07	66.68	58.5	65.78	71.51	70.41	13.18	23.52	46.57
InternVL3-78B	60.03	64.29	52.99	71.97	68.17	71.04	77.18	79.07	57.57	40.89	64.32
Claude-3.5-Sonnet	58.72	56.9	59.58	76.76	70.66	68.38	73.42	76.07	51.01	65.29	65.68
Llama-4-maverick	68.4	62.48	67.37	76.25	69.49	70.93	76.81	78.9	65.29	75.32	71.13
Qwen2-VL-72B	69.13	62.28	64.37	80.34	73.34	69.6	77.58	80.31	65.94	68.48	71.14
GPT-4o	74.2	66.96	70.11	81.17	73.6	69.01	78.95	80.9	67.22	72.31	73.44
Gemini-1.5-Pro	80.01	73.52	78.49	83.33	78.11	75.77	81.28	83.46	69.99	85.7	78.97
Qwen2.5-VL-72B	85.36	76.27	78.16	83.56	79.27	77.81	82.14	83.65	71.09	79.44	79.68

Table 3. Performance of LMMs on subsets of Multilingual OCR track.

Method	Document				Table				Formula		Total
	scan		photo		scan		photo		handwritten		
	en	zh	en	zh	en	zh	en	zh	math	molecular	
Phi-4-Multimodal	15.8	5.47	14.7	3.68	14.09	4.37	4.99	1.44	7.87	12.96	8.54
Cambrian-34B	1.94	12.99	3.47	10.11	0	21.1	15.99	38.89	1.76	12.91	11.87
InternVL2-76B	24.61	16.42	49.42	35.49	64.11	40.63	48.14	30.26	28.33	15.92	35.33
GOT	65.39	69.91	64.07	61.84	47.01*	20.67*	16.90*	11.57*	14.31	20.16	39.18
Claude-3.5-Sonnet	46.85	30.77	60.63	27.39	76.50	49.95	63.67	41.47	43.98	36.65	47.79
Llama-4-maverick	46.97	44.78	50.38	46.82	70.6	55.94	59.88	51.3	46.43	28.18	50.13
GPT-4o	64.19	52.33	65.72	41.32	74.72	57.59	65.89	42.73	41.64	26.83	53.30
Qwen2-VL-72B	66.51	62.36	69.84	60.74	65.96	49.76	59.34	41.15	41.58	20.54	53.78
Gemini-1.5-Pro	64.69	66.01	69.38	68.63	80.99	64.55	65.92	60.26	27.17	56.14	62.37
Qwen2.5-VL-72B	65.19	59.45	64.18	53.13	83.18	74.37	68.07	80.34	52.05	34.4	63.44
InternVL3-78B	68.62	66.42	72.83	69.88	82.5	72.95	71.54	66.84	43.44	21.59	63.66

Table 4. Performance of LMMs on Document Parsing track. * denotes that we convert the outputs of the model to HTML format.

data across different languages.

It is noteworthy that all models perform particularly poorly on handwritten formulas, with scores rarely exceeding 50. The only exceptions are Gemini-1.5 Pro on molecular formula and Qwen2.5-VL-72B on math formula. This indicates that there is significant room for improvement in the recognition of handwritten formulas.

5.4. Key Information Extraction

The KIE task requires the models to generate tokens in structured JSON format, which reduces the number of comparable models to just six generalist models. As presented in Tab. 5, large models demonstrate strong performance in both constrained-category and open-category KIE tasks. Among them, Qwen2.5-VL-72B and InternVL3-78B rank the top two positions, significantly outperforming other models with scores of 89.85% and 77.53%, respectively. This highlights their exceptional capability in structured output generation. We hypothesize that Qwen’s superior performance likely stems from extensive efforts in data curation, potentially encompassing scenarios similar to those in our test set.

5.5. Analysis

In addition to the four main experiments, we conduct additional analyses to uncover the limitations of the current

LMMs. It can be observed that the fine-grained image perception capabilities of current large models are generally weaker compared to their recognition abilities. For example, most models lack grounding capabilities, and their performance tends to decrease when images are rotated.

5.5.1. Visual Text Grounding

We explore the text grounding capabilities of LMMs, which are widely applicable in various scenarios such as image translation, text editing, and text-centric object detection. We evaluate the grounding capabilities using English subsets from the multi-scene OCR track, and the results are presented in Tab. 6. The evaluation metrics are introduced in Appendix. We observe that only five models support text grounding, and their performance in text grounding decreases significantly compared to their text recognition results. Gemini-1.5-Pro achieves the highest score among all the models with a score of 60.98%, while the other models do not exceed 50%. These results demonstrate that fine-grained text grounding remains a major challenge for current LMMs.

5.5.2. Multi-Orientation

Multi-orientation text is a challenge for OCR tasks. Therefore, we evaluate the robustness of large models in handling

Method	Constrained Category				Open Category		Total
	SROIE	CORD	EPHOIE	POIE	SOLD-SIBR	SOLD-HUST-CELL	
Phi-4-Multimodal	35.77	30.94	11.22	29.62	9.23	6.69	20.58
Cambrian-34B	35.91	50.86	2.08	52.52	19.68	21.59	30.44
InternVL2-76B	57.06	66.12	59.03	71.09	50.08	66.22	61.60
Llama-4-maverick	67.75	76.89	35.06	56.04	68.3	72.85	62.81
GPT-4o	72.55	81.26	44.12	59.01	63.60	66.94	63.45
Claude-3.5-Sonnet	69.83	71.51	42.82	59.25	66.20	71.07	64.58
Gemini-1.5-Pro	75.00	78.73	61.27	61.49	59.47	67.72	67.28
Qwen2-VL-72B	75.50	69.02	69.98	62.47	72.43	81.17	71.76
InternVL3-78B	80.69	79.91	71.95	74.33	74.81	83.49	77.53
Qwen2.5-VL-72B	91.14	93.58	88.03	92.93	83.59	89.82	89.85

Table 5. Results of LMMs on subsets of KIE track.

Method	Natural Scene OCR				Document OCR			UGC Image OCR	Total
	TotalText	IC15	InverseText	HierAgent	FUNSD	CORD	IAM	LAION-OCR	
KOSMOS2.5	17.6	28.09	8.55	20.64	13.59	31.59	25.12	25.12	21.29
Florence	27.02	31.38	22.49	23.35	8.85	43.52	0.2	37.09	24.24
Qwen2.5-VL-72B	52.64	56.99	45.19	24.98	32.25	40.98	1.44	71.41	40.74
TextMonkey	58.87	35.17	38.15	31.37	9.42	69.94	58.39	73.71	46.95
Gemini-1.5-Pro	67.25	52.00	66.77	47.74	43.82	76.48	56.05	77.71	60.98

Table 6. Comparison of text grounding performance on English subsets from Multi-Scene OCR.

Method	Total		Diff
	Normal	M.O.	
GOT	61.00	26.14	-34.86
KOSMOS2.5	47.55	28.84	-18.71
InternVL2-76B	76.92	38.00	-38.92
Claude-3.5-Sonnet	72.87	42.42	-30.45
Qwen2-VL-72B	77.95	54.62	-23.33
GPT-4o	76.40	57.02	-19.38
Qwen2.5-VL-72B	86.40	73.13	-13.27
Gemini-1.5-Pro	83.25	79.45	-3.80

Method	$R_{rep}(\downarrow)$
InternVL2-76B	5.94%
Qwen2-VL-72B	0.65%
Qwen2.5-VL-72B	0.35%
GPT-4o	0.28%
Gemini-1.5-Pro	0.14%
Claude-3.5-Sonnet	0.09%
TextMonkey	33.93%
KOSMOS2.5	10.64%
GOT	2.72%
Florence	0.68%

Table 7. Comparison of results on the normal and multi-oriented benchmark of Multi-Scene OCR task with English and Chinese. Note that M.O. stands for multi-oriented benchmark.

multi-orientation text, and the experimental results are presented in Tab. 7. We randomly rotate the images of the multi-scene OCR benchmark by 0, 90, 180, or 270 degrees to build a multi-orientation benchmark. It can be observed that most methods perform much worse on multi-orientation images than on original images. Claude-3.5-Sonnet, GOT, Qwen2-VL-72B, and InternVL2 show a performance decrease of more than 23%. GPT-4o, KOSMOS2.5 and Qwen2.5-VL-72B result in reduction more than 13%. While Gemini-1.5-Pro exhibits only a slight decline of 3.80%. These results emphasize the need for the inclusion of more rotated training data to mitigate this challenge.

5.5.3. Hallucination of Repetition Output

Based on our observation, most large models tend to produce hallucinations in the form of repeated text during prediction. So we devise a repetition ratio R_{rep} by measuring the number of images producing repetitive output out of the overall number of images. Detailed introduction of R_{rep} can be referred to Appendix. The repetition performance is presented in Tab. 8. We observe that generalist LMMs produce less repetition results than specialized methods. Among all the models, Claude-3.5-Sonnet has the lowest repetition rate,

Table 8. Hallucination of repetition of LMMs on data from Multi-Scene OCR track. R_{rep} denotes the rate of Repetition.

with only one in a thousand images exhibiting repeated outputs. TextMonkey, on the other hand, has the most severe hallucination of repetition, with a rate of 33.93%.

6. Conclusions and Future Works

In this paper, we introduce CC-OCR, a comprehensive and challenging benchmark designed to evaluate the visual literacy and document processing capabilities of state-of-the-art LMMs. By covering a diverse range of scenarios, tasks, and challenges, CC-OCR addresses critical gaps in existing benchmarks, offering a holistic evaluation framework for multilingual text reading, multi-scene OCR, document parsing, and key information extraction. Through rigorous experiments on 16 prominent models, including ten generalist, four specialized LMMs and two OCR APIs, we underscore the importance of handling fine-grained text grounding, multi-oriented and multilingual text, handwritten formulas, and hallucinations of repetition errors to advance the literate intelligence of LMMs. We hope that CC-OCR will serve as a valuable and challenging benchmark for the research community, driving progress in the development of more capable and reliable LMMs for diverse real-world OCR tasks.

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